

VECTORS



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Vectors

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1.1: Prelude to Vectors

Vectors are essential to physics and engineering. Many fundamental physical quantities are vectors, including displacement, velocity, force, and electric and magnetic vector fields. Scalar products of vectors define other fundamental scalar physical quantities, such as energy. Vector products of vectors define still other fundamental vector physical quantities, such as torque and angular momentum. In other words, vectors are a component part of physics in much the same way as sentences are a component part of literature.



Figure 1.1.1: A signpost gives information about distances and directions to towns or to other locations relative to the location of the signpost. Distance is a scalar quantity. Knowing the distance alone is not enough to get to the town; we must also know the direction from the signpost to the town. The direction, together with the distance, is a vector quantity commonly called the displacement vector. A signpost, therefore, gives information about displacement vectors from the signpost to towns. (credit: modification of work by “studio tdes”/Flickr)

In introductory physics, vectors are Euclidean quantities that have geometric representations as arrows in one dimension (in a line), in two dimensions (in a plane), or in three dimensions (in space). They can be added, subtracted, or multiplied. In this chapter, we explore elements of vector algebra for applications in mechanics and in electricity and magnetism. Vector operations also have numerous generalizations in other branches of physics.

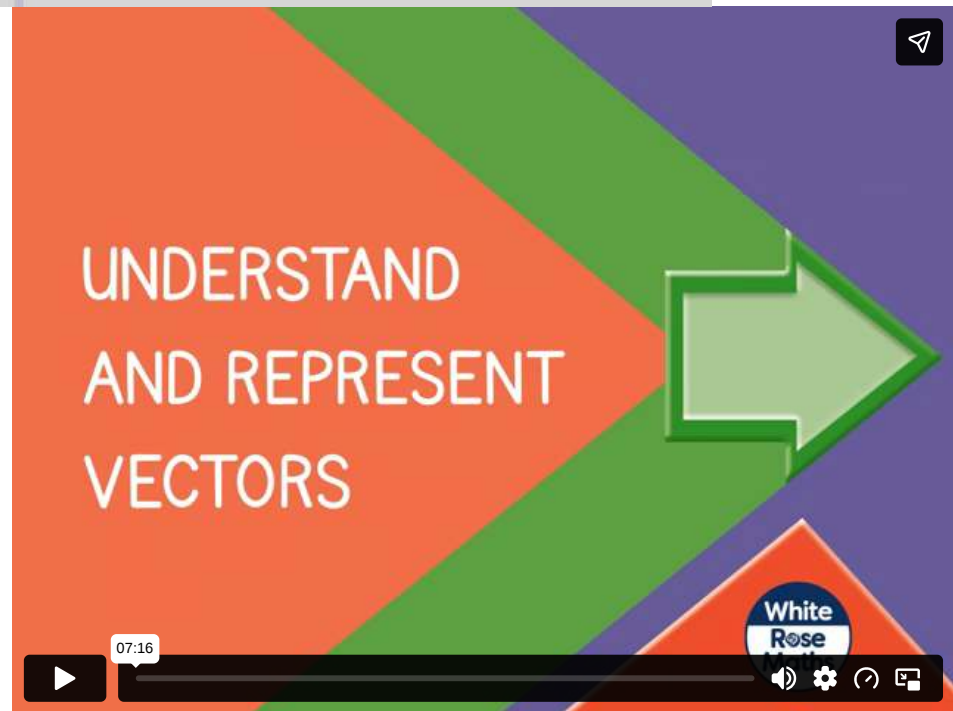
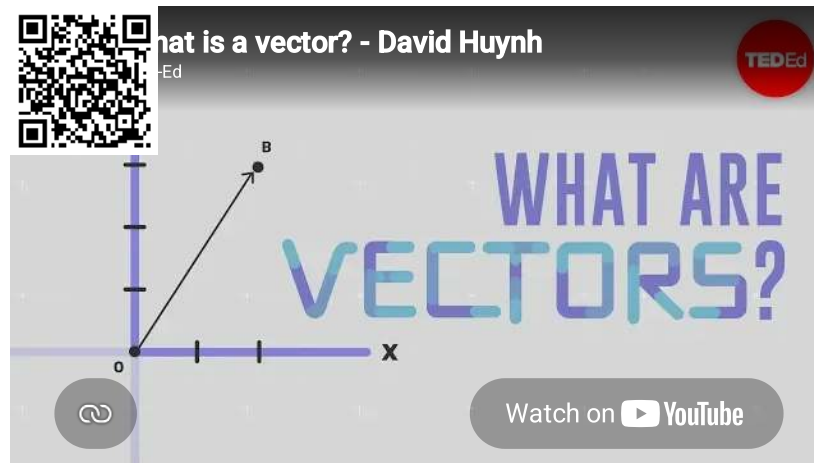
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1.2: Scalars and Vectors Part 1

To better understand the science of propulsion it is necessary to use some mathematical ideas from **vector analysis**. Most people are introduced to vectors in high school or college, but for those who were not introduced let's start with some videos

DON'T PANIC!



Spr10.3.1 - Understand and represent vectors from White Rose Education on Vimeo.

There are many complex parts to vector analysis and we aren't going there. We are going to limit ourselves to the very basics. Vectors allow us to look at complex, multi-dimensional problems as a simpler group of one-dimensional problems. We will be concerned mostly with definitions. The words are a bit strange, but the ideas are very powerful as you will see. If you want to find out a lot more about vectors you can download this report on [vector analysis](#).

Math and science were invented by humans to describe and understand the world around us. We live in a (at least) four-dimensional world governed by the passing of time and three space dimensions; up and down, left and right, and back and forth. We observe that there are some quantities and processes in our world that depend on the **direction** in which they occur, and there are some quantities that do not depend on direction. For example, the **volume** of an object, the three-dimensional space that an object occupies, does not depend on direction. If we have a 5 cubic foot block of iron and we move it up and down and then left and right, we still have a 5 cubic foot block of iron. On the other hand, the **location**, of an object does depend on direction. If we move the 5 cubic foot block 5 miles to the north, the resulting location is very different than if we moved it 5 miles to the east. Mathematicians and scientists call a quantity which depends on direction a **vector quantity**. A quantity which does not depend on direction is called a **scalar quantity**.

Vector quantities have two characteristics, a magnitude and a direction. Scalar quantities have only a magnitude. When **comparing** two vector quantities of the same type, you have to compare both the magnitude and the direction. For scalars, you only have to **compare** the magnitude. When doing any mathematical operation on a vector quantity (like adding, subtracting, multiplying ..) you have to **consider** both the magnitude and the direction. This makes dealing with vector quantities a little more complicated than scalars.

On the slide we list some of the physical quantities discussed in the [Beginner's Guide to Aeronautics](#) and group them into either vector or scalar quantities. Of particular interest, the **forces** which operate on a flying aircraft, the **weight**, **thrust**, and **aerodynamic forces**, are all vector quantities. The resulting **motion** of the aircraft in terms of displacement, velocity, and acceleration are also vector quantities. These quantities can be determined by application of [Newton's laws](#) for vectors. The scalar quantities include most of the

[thermodynamic state](#) variables involved with the propulsion system, such as the [density](#), [pressure](#), and [temperature](#) of the propellants. The [energy](#), [work](#), and [entropy](#) associated with the engines are also scalar quantities.

Vectors have magnitude and direction, scalars only have magnitude. The fact that **magnitude** occurs for both scalars and vectors can lead to some confusion. There are some quantities, like **speed**, which have very special definitions for scientists. By definition, speed is the scalar magnitude of a **velocity** vector. A car going down the road has a speed of 50 mph. Its velocity is 50 mph in the northeast direction. It can get very confusing when the terms are used interchangeably! Another example is **mass** and **weight**. Weight is a force which is a vector and has a magnitude and direction. Mass is a scalar. [Weight and mass](#) are related to one another, but they are not the same quantity.`

While Newton's laws describe the resulting motion of a solid, there are special equations which describe the motion of fluids, [gases and liquids](#). For any physical system, the [mass](#), [momentum](#), and [energy](#) of the system must be conserved. Mass and energy are scalar quantities, while momentum is a vector quantity. This results in a **coupled** set of equations, called the [Navier-Stokes equations](#), which describe how fluids behave when subjected to external forces. These equations are the fluid equivalent of Newton's laws of motion and are very difficult to solve and understand. A simplified version of the equations called the [Euler equations](#) can be solved for some fluids problems.

Here are some examples of scalar and vector quantities

Scalar Quantities	Vector Quantities
Length, area, volume	Displacement
Speed	Velocity
Mass, density	Acceleration
Pressure	Momentum
Temperature	Force
Energy	Lift, drag, thrust
Entropy	
Work, power	

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1.3: Scalars and Vectors Part 2

Learning Objectives

- Describe the difference between vector and scalar quantities.
- Identify the magnitude and direction of a vector.
- Explain the effect of multiplying a vector quantity by a scalar.
- Describe how one-dimensional vector quantities are added or subtracted.
- Explain the geometric construction for the addition or subtraction of vectors in a plane.
- Distinguish between a vector equation and a scalar equation.

Many familiar physical quantities can be specified completely by giving a single number and the appropriate unit. For example, “a class period lasts 50 min” or “the gas tank in my car holds 65 L” or “the distance between two posts is 100 m.” A physical quantity that can be specified completely in this manner is called a **scalar quantity**. Scalar is a synonym of “number.” Time, mass, distance, length, volume, temperature, and energy are examples of **scalar** quantities.

Scalar quantities that have the same physical units can be added or subtracted according to the usual rules of algebra for numbers. For example, a class ending 10 min earlier than 50 min lasts $50 \text{ min} - 10 \text{ min} = 40 \text{ min}$. Similarly, a 60-cal serving of corn followed by a 200-cal serving of donuts gives $60 \text{ cal} + 200 \text{ cal} = 260 \text{ cal}$ of energy. When we multiply a scalar quantity by a number, we obtain the same scalar quantity but with a larger (or smaller) value. For example, if yesterday’s breakfast had 200 cal of energy and today’s breakfast has four times as much energy as it had yesterday, then today’s breakfast has $4(200 \text{ cal}) = 800 \text{ cal}$ of energy. Two scalar quantities can also be multiplied or divided by each other to form a derived scalar quantity. For example, if a train covers a distance of 100 km in 1.0 h, its speed is $100.0 \text{ km}/1.0 \text{ h} = 27.8 \text{ m/s}$, where the speed is a derived scalar quantity obtained by dividing distance by time.

Many physical quantities, however, cannot be described completely by just a single number of physical units. For example, when the U.S. Coast Guard dispatches a ship or a helicopter for a rescue mission, the rescue team must know not only the distance to the distress signal, but also the direction from which the signal is coming so they can get to its origin as quickly as possible. Physical quantities specified completely by giving a number of units (magnitude) and a direction are called **vector quantities**. Examples of vector quantities include displacement, velocity, position, force, and torque. In the language of mathematics, physical vector quantities are represented by mathematical objects called **vectors** (Figure 1.3.1). We can add or subtract two vectors, and we can multiply a vector by a scalar or by another vector, but we cannot divide by a vector. The operation of division by a vector is not defined.

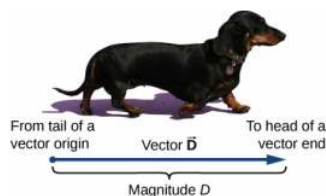


Figure 1.3.1: We draw a vector from the initial point or origin (called the “tail” of a vector) to the end or terminal point (called the “head” of a vector), marked by an arrowhead. Magnitude is the length of a vector and is always a positive scalar quantity. (credit: modification of work by Cate Sevilla)

Let’s examine vector algebra using a graphical method to be aware of basic terms and to develop a qualitative understanding. In practice, however, when it comes to solving physics problems, we use analytical methods, which we’ll see in the next section. Analytical methods are more simple computationally and more accurate than graphical methods. From now on, to distinguish between a vector and a scalar quantity, we adopt the common convention that a letter in bold type with an arrow above it denotes a vector, and a letter without an arrow denotes a scalar. For example, a distance of 2.0 km, which is a scalar quantity, is denoted by $d = 2.0 \text{ km}$, whereas a displacement of 2.0 km in some direction, which is a vector quantity, is denoted by $\vec{d}$.

Suppose you tell a friend on a camping trip that you have discovered a terrific fishing hole 6 km from your tent. It is unlikely your friend would be able to find the hole easily unless you also communicate the direction in which it can be found with respect to your campsite. You may say, for example, “Walk about 6 km northeast from my tent.” The key concept here is that you have to give not one but two pieces of information—namely, the distance or magnitude (6 km) **and** the direction (northeast).

Displacement is a general term used to describe a change in position, such as during a trip from the tent to the fishing hole. Displacement is an example of a vector quantity. If you walk from the tent (location A) to the hole (location B), as shown in Figure 1.3.2, the vector $\vec{D}$, representing your **displacement**, is drawn as the arrow that originates at point A and ends at point B. The arrowhead marks the end of the vector. The direction of the displacement vector $\vec{D}$ is the direction of the arrow. The length of the arrow represents the **magnitude** D of vector $\vec{D}$. Here, $D = 6$ km. Since the magnitude of a vector is its length, which is a positive number, the magnitude is also indicated by placing the absolute value notation around the symbol that denotes the vector; so, we can write equivalently that $D = |\vec{D}|$. To solve a vector problem graphically, we need to draw the vector $\vec{D}$ to scale. For example, if we assume 1 unit of distance (1 km) is represented in the drawing by a line segment of length $u = 2$ cm, then the total displacement in this example is represented by a vector of length $d = 6u = 6(2 \text{ cm}) = 12 \text{ cm}$, as shown in Figure 1.3.3. Notice that here, to avoid confusion, we used $D = 6$ km to denote the magnitude of the actual displacement and $d = 12$ cm to denote the length of its representation in the drawing.

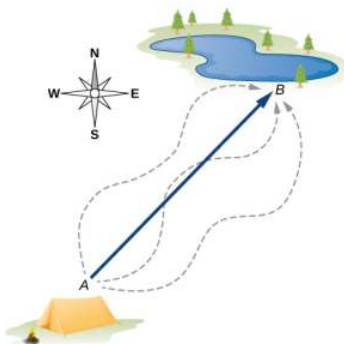


Figure 1.3.2: The displacement vector from point A (the initial position at the campsite) to point B (the final position at the fishing hole) is indicated by an arrow with origin at point A and end at point B. The displacement is the same for any of the actual paths (dashed curves) that may be taken between points A and B.

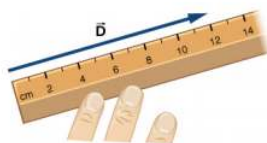


Figure 1.3.3: A displacement $\vec{D}$ of magnitude 6 km is drawn to scale as a vector of length 12 cm when the length of 2 cm represents 1 unit of displacement (which in this case is 1 km).

Suppose your friend walks from the campsite at A to the fishing pond at B and then walks back: from the fishing pond at B to the campsite at A. The magnitude of the displacement vector $\vec{D}_{AB}$ from A to B is the same as the magnitude of the displacement vector $\vec{D}_{BA}$ from B to A (it equals 6 km in both cases), so we can write $\vec{D}_{AB} = \vec{D}_{BA}$. However, vector $\vec{D}_{AB}$ is not equal to vector $\vec{D}_{BA}$ because these two vectors have different directions: $\vec{D}_{AB} \neq \vec{D}_{BA}$. In Figure 2.3, vector $\vec{D}_{BA}$ would be represented by a vector with an origin at point B and an end at point A, indicating vector $\vec{D}_{BA}$ points to the southwest, which is exactly 180° opposite to the direction of vector $\vec{D}_{AB}$. We say that vector $\vec{D}_{BA}$ is **antiparallel** to vector $\vec{D}_{AB}$ and write $\vec{D}_{AB} = -\vec{D}_{BA}$, where the minus sign indicates the antiparallel direction.

Two vectors that have identical directions are said to be **parallel vectors**—meaning, they are **parallel** to each other. Two parallel vectors $\vec{A}$ and $\vec{B}$ are equal, denoted by $\vec{A} = \vec{B}$, if and only if they have equal magnitudes $|\vec{A}| = |\vec{B}|$. Two vectors with directions perpendicular to each other are said to be **orthogonal vectors**. These relations between vectors are illustrated in Figure 1.3.4.

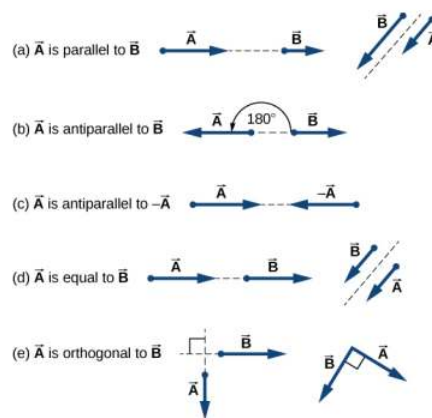


Figure 1.3.4: Various relations between two vectors $\vec{A}$ and $\vec{B}$. (a) $\vec{A} \neq \vec{B}$ because $A \neq B$. (b) $\vec{A} \neq \vec{B}$ because they are not parallel and $A \neq B$. (c) $\vec{A} \neq -\vec{A}$ because they have different directions (even though $|\vec{A}| = |-\vec{A}| = A$). (d) $\vec{A} = \vec{B}$ because they are parallel and have identical magnitudes $A = B$. (e) $\vec{A} \neq \vec{B}$ because they have different directions (are not parallel); here, their directions differ by 90° —meaning, they are orthogonal.

? Exercise 2.1

Two motorboats named **Alice** and **Bob** are moving on a lake. Given the information about their velocity vectors in each of the following situations, indicate whether their velocity vectors are equal or otherwise.

- Alice** moves north at 6 knots and **Bob** moves west at 6 knots.
- Alice** moves west at 6 knots and **Bob** moves west at 3 knots.
- Alice** moves northeast at 6 knots and **Bob** moves south at 3 knots.
- Alice** moves northeast at 6 knots and **Bob** moves southwest at 6 knots.
- Alice** moves northeast at 2 knots and **Bob** moves closer to the shore northeast at 2 knots.

Algebra of Vectors in One Dimension

Vectors can be multiplied by scalars, added to other vectors, or subtracted from other vectors. We can illustrate these vector concepts using an example of the fishing trip seen in Figure 1.3.5.

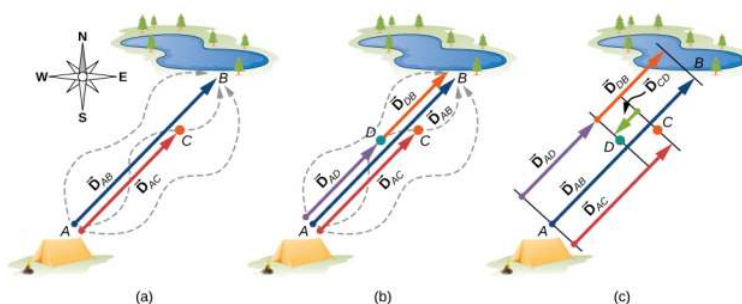


Figure 1.3.5: Displacement vectors for a fishing trip. (a) Stopping to rest at point C while walking from camp (point A) to the pond (point B). (b) Going back for the dropped tackle box (point D). (c) Finishing up at the fishing pond.

Suppose your friend departs from point A (the campsite) and walks in the direction to point B (the fishing pond), but, along the way, stops to rest at some point C located three-quarters of the distance between A and B, beginning from point A (Figure 1.3.5a). What is his displacement vector $\vec{D}_{AC}$ when he reaches point C? We know that if he walks all the way to B, his displacement vector relative to A is $\vec{D}_{AB}$, which has magnitude $D_{AB} = 6$ km and a direction of northeasterly. If he walks only a 0.75 fraction of the total distance, maintaining the northeasterly direction, at point C he must be $0.75 D_{AB} = 4.5$ km away from the campsite at A. So, his displacement vector at the rest point C has magnitude $D_{AC} = 4.5$ km $= 0.75 D_{AB}$ and is parallel to the displacement vector $\vec{D}_{AB}$. All of this can be stated succinctly in the form of the following **vector equation**:

$$\vec{D}_{AC} = 0.75 \vec{D}_{AB}.$$

In a vector equation, both sides of the equation are vectors. The previous equation is an example of a vector multiplied by a positive scalar (number) $\alpha = 0.75$. The result, $\vec{D}_{AC}$, of such a multiplication is a new vector with a direction parallel to the direction of the original vector $\vec{D}_{AB}$. In general, when a vector $\vec{D}_A$ is multiplied by a positive scalar α , the result is a new vector $\vec{D}_B$ that is parallel to $\vec{D}_A$:

$$\vec{B} = \alpha \vec{A} \quad (1.3.1)$$

The magnitude $|\vec{B}|$ of this new vector is obtained by multiplying the magnitude $|\vec{A}|$ of the original vector, as expressed by the **scalar equation**:

$$B = |\alpha|A. \quad (1.3.2)$$

In a scalar equation, both sides of the equation are numbers. Equation 1.3.2 is a scalar equation because the magnitudes of vectors are scalar quantities (and positive numbers). If the scalar α is **negative** in the vector equation Equation 1.3.1, then the magnitude $|\vec{B}|$ of the new vector is still given by Equation 1.3.2, but the direction of the new vector $\vec{B}$ is **antiparallel** to the direction of $\vec{A}$. These principles are illustrated in Figure 1.3.6a by two examples where the length of vector $\vec{A}$ is 1.5 units. When $\alpha = 2$, the new vector $\vec{B} = 2\vec{A}$ has length $B = 2A = 3.0$ units (twice as long as the original vector) and is parallel to the original vector. When $\alpha = -2$, the new vector $\vec{C} = -2\vec{A}$ has length $C = |-2|A = 3.0$ units (twice as long as the original vector) and is antiparallel to the original vector.

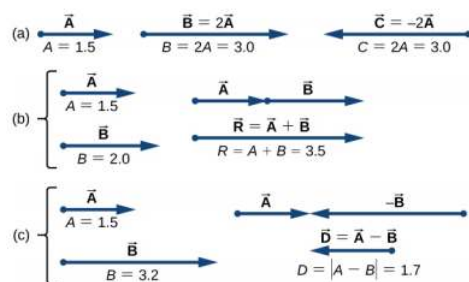


Figure 1.3.6: Algebra of vectors in one dimension. (a) Multiplication by a scalar. (b) Addition of two vectors ($\vec{R}$ is called the resultant of vectors ($\vec{A}$ and $\vec{B}$)). (c) Subtraction of two vectors ($\vec{D}$ is the difference of vectors ($\vec{A}$ and $\vec{B}$)).

Now suppose your fishing buddy departs from point A (the campsite), walking in the direction to point B (the fishing hole), but he realizes he lost his tackle box when he stopped to rest at point C (located three-quarters of the distance between A and B, beginning from point A). So, he turns back and retraces his steps in the direction toward the campsite and finds the box lying on the path at some point D only 1.2 km away from point C (see Figure 1.3.5b). What is his displacement vector $\vec{D}_{AD}$ when he finds the box at point D? What is his displacement vector $\vec{D}_{DB}$ from point D to the hole? We have already established that at rest point C his displacement vector is $\vec{D}_{AC} = 0.75 \vec{D}_{AB}$. Starting at point C, he walks southwest (toward the campsite), which means his new displacement vector $\vec{D}_{CD}$ from point C to point D is antiparallel to $\vec{D}_{AB}$. Its magnitude $|\vec{D}_{CD}|$ is $D_{CD} = 1.2 \text{ km} = 0.2 D_{AB}$, so his second displacement vector is $\vec{D}_{CD} = -0.2 \vec{D}_{AB}$. His total displacement $\vec{D}_{AD}$ relative to the campsite is the vector sum of the two displacement vectors: vector $\vec{D}_{AC}$ (from the campsite to the rest point) and vector $\vec{D}_{CD}$ (from the rest point to the point where he finds his box):

$$\vec{D}_{AD} = \vec{D}_{AC} + \vec{D}_{CD}. \quad (1.3.3)$$

The vector sum of two (or more vectors) is called the **resultant vector** or, for short, the **resultant**. When the vectors on the right-hand-side of Equation 1.3.3 are known, we can find the resultant $\vec{D}_{AD}$ as follows:

$$\vec{D}_{AD} = \vec{D}_{AC} + \vec{D}_{CD} = 0.75 \vec{D}_{AB} - 0.2 \vec{D}_{AB} = (0.75 - 0.2) \vec{D}_{AB} = 0.55 \vec{D}_{AB}. \quad (1.3.4)$$

When your friend finally reaches the pond at B, his displacement vector $\vec{D}_{AB}$ from point A is the vector sum of his displacement vector $\vec{D}_{AD}$ from point A to point D and his displacement vector $\vec{D}_{DB}$ from point D to the fishing hole: $\vec{D}_{AB} = \vec{D}_{AD} + \vec{D}_{DB}$ (see Figure 1.3.5c). This means his displacement vector $\vec{D}_{DB}$ is the difference of two vectors:

$$\vec{D}_{DB} = \vec{D}_{AB} - \vec{D}_{AD} = \vec{D}_{AB} + (-\vec{D}_{AD}). \quad (1.3.5)$$

Notice that a difference of two vectors is nothing more than a vector sum of two vectors because the second term in Equation 1.3.5 is vector $-\vec{D}_{AD}$ (which is antiparallel to $\vec{D}_{AD}$). When we substitute Equation 1.3.4 into Equation 1.3.5, we obtain the second displacement vector:

$$D_{DB} = D_{AB} - D_{AD} = D_{AB} - 0.55 D_{AB} = (1.0 - 0.55) D_{AB} = 0.45 D_{AB}. \quad (1.3.6)$$

This result means your friend walked $D_{DB} = 0.45 D_{AB} = 0.45(6.0 \text{ km}) = 2.7 \text{ km}$ from the point where he finds his tackle box to the fishing hole.

When vectors $\vec{A}$ and $\vec{B}$ lie along a line (that is, in one dimension), such as in the camping example, their resultant $\vec{R} = \vec{A} + \vec{B}$ and their difference $\vec{D} = \vec{A} - \vec{B}$ both lie along the same direction. We can illustrate the addition or subtraction of vectors by drawing the corresponding vectors to scale in one dimension, as shown in Figure 1.3.6.

To illustrate the resultant when $\vec{A}$ and $\vec{B}$ are two parallel vectors, we draw them along one line by placing the origin of one vector at the end of the other vector in head-to-tail fashion (see Figure (\PageIndex{6b})). The magnitude of this resultant is the sum of their magnitudes: $R = A + B$. The direction of the resultant is parallel to both vectors. When vector $\vec{A}$ is antiparallel to vector $\vec{B}$, we draw them along one line in either head-to-head fashion (Figure (\PageIndex{6c})) or tail-to-tail fashion. The magnitude of the vector difference, then, is the **absolute value** $D = |A - B|$ of the difference of their magnitudes. The direction of the difference vector $\vec{D}$ is parallel to the direction of the longer vector.

In general, in one dimension—as well as in higher dimensions, such as in a plane or in space—we can add any number of vectors and we can do so in any order because the addition of vectors is **commutative**,

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}. \quad (1.3.7)$$

and **associative**,

$$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C}). \quad (1.3.8)$$

Moreover, multiplication by a scalar is **distributive**:

$$\alpha_1 \vec{A} + \alpha_2 \vec{A} = (\alpha_1 + \alpha_2) \vec{A}. \quad (1.3.9)$$

We used the distributive property in Equation 1.3.4 and Equation 1.3.6.

When adding many vectors in one dimension, it is convenient to use the concept of a **unit vector**. A unit vector, which is denoted by a letter symbol with a hat, such as $\hat{u}$, has a magnitude of one and does not have any physical unit so that $|\hat{u}| \equiv u = 1$. The only role of a unit vector is to specify direction. For example, instead of saying vector $\vec{D}_{AB}$ has a magnitude of 6.0 km and a direction of northeast, we can introduce a unit vector $\hat{u}$ that points to the northeast and say succinctly that $\vec{D}_{AB} = (6.0 \text{ km}) \hat{u}$. Then the southwesterly direction is simply given by the unit vector $-\hat{u}$. In this way, the displacement of 6.0 km in the southwesterly direction is expressed by the vector

$$\vec{D}_{BA} = (-6.0 \text{ km}) \hat{u}.$$

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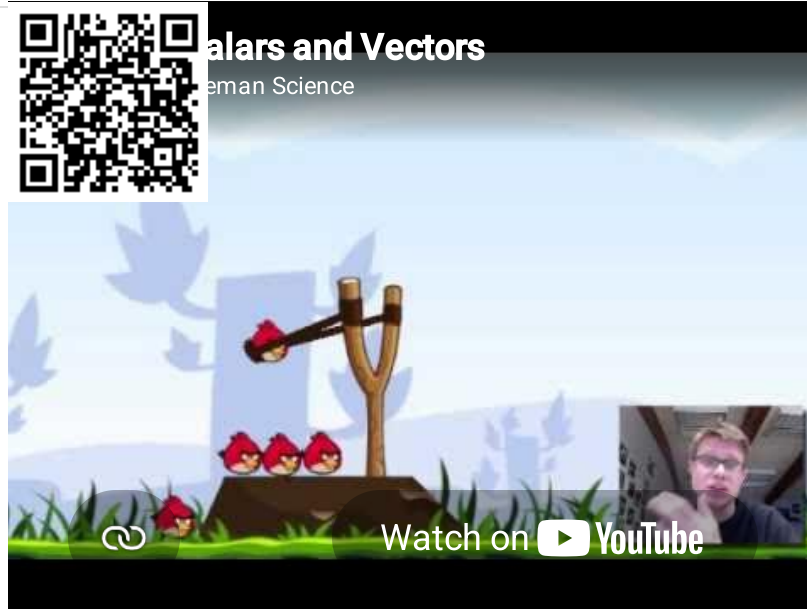
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1.4: One Dimensional Motion (Video) Khan Academy



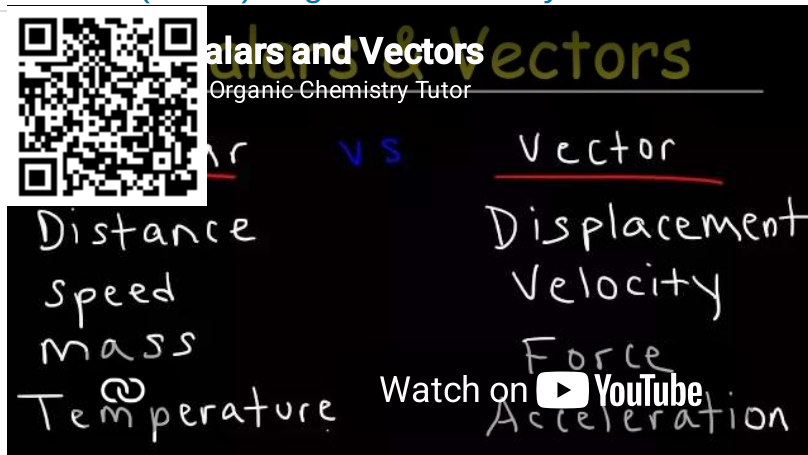
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1.7: Additional Materials

[Scalar and Vector Quantities](#) NASA

Link: [https://www.grc.nasa.gov/www/k-](https://www.grc.nasa.gov/www/k-12/airplane/vectors.html#:~:text=Mathematicians%20and%20scientists%20call%20a,quantities%20have%20only%20a%20magnitude.)

[12/airplane/vectors.html#:~:text=Mathematicians%20and%20scientists%20call%20a,quantities%20have%20only%20a%20magnitude.](#)

[Scalars and Vectors](#) Physics Classroom

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CHAPTER OVERVIEW

2: Addition of Vectors- Graphical Method

[2.1: Vector Addition in Two Dimensions](#)

[2.2: Graphical Methods of Vector Addition](#)

[2.3: Adding Vectors \(Algebraically and Graphically\) \(Video\) Khan Academy](#)

[2.4: Adding Two Vectors Graphically \(Video\) Math and Science](#)

[2.5: Adding Vectors Graphically \(Video\) Michel van Biezen](#)

[2.6: Additional Materials](#)

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2.1: Vector Addition in Two Dimensions

✓ Example 2.1.1: A Ladybug Walker

A long measuring stick rests against a wall in a physics laboratory with its 200-cm end at the floor. A ladybug lands on the 100-cm mark and crawls randomly along the stick. It first walks 15 cm toward the floor, then it walks 56 cm toward the wall, then it walks 3 cm toward the floor again. Then, after a brief stop, it continues for 25 cm toward the floor and then, again, it crawls up 19 cm toward the wall before coming to a complete rest (Figure 2.1.1). Find the vector of its total displacement and its final resting position on the stick.

Strategy

If we choose the direction along the stick toward the floor as the direction of unit vector $\hat{u}$, then the direction toward the floor is $+\hat{u}$ and the direction toward the wall is $-\hat{u}$. The ladybug makes a total of five displacements:

$$\begin{aligned}\vec{D}_1 &= (15 \text{ cm})(+\hat{u}), \\ \vec{D}_2 &= (56 \text{ cm})(-\hat{u}), \\ \vec{D}_3 &= (3 \text{ cm})(+\hat{u}), \\ \vec{D}_4 &= (25 \text{ cm})(+\hat{u}), \text{ and} \\ \vec{D}_5 &= (19 \text{ cm})(-\hat{u}).\end{aligned}$$

The total displacement $\vec{D}$ is the resultant of all its displacement vectors.

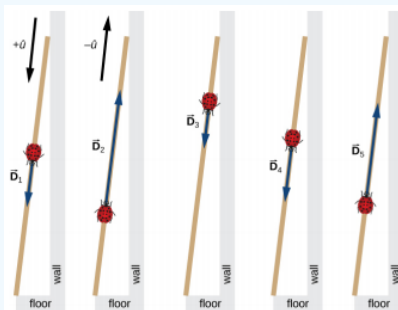


Figure 2.1.1: Five displacements of the ladybug. Note that in this schematic drawing, magnitudes of displacements are not drawn to scale. (credit: modification of work by “Persian Poet Gal”/Wikimedia Commons)

Solution

The resultant of all the displacement vectors is

$$\begin{aligned}\vec{D} &= \vec{D}_1 + \vec{D}_2 + \vec{D}_3 + \vec{D}_4 + \vec{D}_5 \\ &= (15 \text{ cm})(+\hat{u}) + (56 \text{ cm})(-\hat{u}) + (3 \text{ cm})(+\hat{u}) + (25 \text{ cm})(+\hat{u}) + (19 \text{ cm})(-\hat{u}) \\ &= (15 - 56 + 3 + 25 - 19) \text{ cm } \hat{u} \\ &= -32 \text{ cm } \hat{u}.\end{aligned}$$

In this calculation, we use the distributive law given by [Equation 2.2.9](#). The result reads that the total displacement vector points away from the 100-cm mark (initial landing site) toward the end of the meter stick that touches the wall. The end that touches the wall is marked 0 cm, so the final position of the ladybug is at the $(100 - 32) \text{ cm} = 68\text{-cm}$ mark.

? Exercise 2.2

A cave diver enters a long underwater tunnel. When her displacement with respect to the entry point is 20 m, she accidentally drops her camera, but she doesn't notice it missing until she is some 6 m farther into the tunnel. She swims back 10 m but cannot find the camera, so she decides to end the dive. How far from the entry point is she? Taking the positive direction out of the tunnel, what is her displacement vector relative to the entry point?

Algebra of Vectors in Two Dimensions

When vectors lie in a plane—that is, when they are in two dimensions—they can be multiplied by scalars, added to other vectors, or subtracted from other vectors in accordance with the general laws expressed by [Equation 2.2.1](#), [Equation 2.2.2](#), [Equation 2.2.7](#), and [Equation 2.2.8](#). However, the addition rule for two vectors in a plane becomes more complicated than the rule for vector addition in one dimension. We have to use the laws of geometry to construct resultant vectors, followed by trigonometry to find vector magnitudes and directions. This geometric approach is commonly used in navigation (Figure 2.1.2). In this section, we need to have at hand two rulers, a triangle, a protractor, a pencil, and an eraser for drawing vectors to scale by geometric constructions.



Figure 2.1.2: In navigation, the laws of geometry are used to draw resultant displacements on nautical maps.

For a geometric construction of the sum of two vectors in a plane, we follow the **parallelogram rule**. Suppose two vectors $\vec{A}$ and $\vec{B}$ are at the arbitrary positions shown in Figure 2.1.3. Translate either one of them in parallel to the beginning of the other vector, so that after the translation, both vectors have their origins at the same point. Now, at the end of vector $\vec{A}$ we draw a line parallel to vector $\vec{B}$ and at the end of vector $\vec{B}$ we draw a line parallel to vector $\vec{A}$ (the dashed lines in Figure 2.1.3). In this way, we obtain a parallelogram. From the origin of the two vectors we draw a diagonal that is the resultant $\vec{R}$ of the two vectors: $\vec{R} = \vec{A} + \vec{B}$ (Figure 2.1.3a). The other diagonal of this parallelogram is the vector difference of the two vectors $\vec{D} = \vec{A} - \vec{B}$, as shown in Figure 2.1.3b. Notice that the end of the difference vector is placed at the end of vector $\vec{A}$.

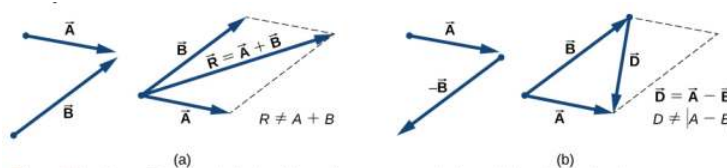


Figure 2.1.3: The parallelogram rule for the addition of two vectors. Make the parallel translation of each vector to a point where their origins (marked by the dot) coincide and construct a parallelogram with two sides on the vectors and the other two sides (indicated by dashed lines) parallel to the vectors. (a) Draw the resultant vector $\vec{R}$ along the diagonal of the parallelogram from the common point to the opposite corner. Length R of the resultant vector is not equal to the sum of the magnitudes of the two vectors. (b) Draw the difference vector $\vec{D} = \vec{A} - \vec{B}$ along the diagonal connecting the ends of the vectors. Place the origin of vector $\vec{D}$ at the end of vector $\vec{B}$ and the end (arrowhead) of vector $\vec{D}$ at the end of vector $\vec{A}$. Length D of the difference vector is not equal to the difference of magnitudes of the two vectors.

It follows from the parallelogram rule that neither the magnitude of the resultant vector nor the magnitude of the difference vector can be expressed as a simple sum or difference of magnitudes A and B , because the length of a diagonal cannot be expressed as a simple sum of side lengths. When using a geometric construction to find magnitudes $|\vec{R}|$ and $|\vec{D}|$, we have to use trigonometry laws for triangles, which may lead to complicated algebra. There are two ways to circumvent this algebraic complexity. One way is to use the method of components, which we examine in the next section. The other way is to draw the vectors to scale, as is done in navigation, and read approximate vector lengths and angles (directions) from the graphs. In this section we examine the second approach.

If we need to add three or more vectors, we repeat the parallelogram rule for the pairs of vectors until we find the resultant of all of the resultants. For three vectors, for example, we first find the resultant of vector 1 and vector 2, and then we find the resultant of this resultant and vector 3. The order in which we select the pairs of vectors does not matter because the operation of vector addition is commutative and associative (see [Equation 2.2.7](#) and [Equation 2.2.8](#)). Before we state a general rule that follows from repetitive applications of the parallelogram rule, let's look at the following example.

Suppose you plan a vacation trip in Florida. Departing from Tallahassee, the state capital, you plan to visit your uncle Joe in Jacksonville, see your cousin Vinny in Daytona Beach, stop for a little fun in Orlando, see a circus performance in Tampa, and visit the University of Florida in Gainesville. Your route may be represented by five displacement vectors $\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$, and $\vec{E}$, which are indicated by the red vectors in Figure 2.1.4. What is your total displacement when you reach Gainesville? The total displacement is the vector sum of all five displacement vectors, which may be found by using the parallelogram rule four times. Alternatively, recall that the displacement vector has its beginning at the initial position (Tallahassee) and its end at the final position (Gainesville), so the total displacement vector can be drawn directly as an arrow connecting Tallahassee with Gainesville (see the green vector in Figure 2.1.4). When we use the parallelogram rule four times, the resultant $\vec{R}$ we obtain is exactly this green vector connecting Tallahassee with Gainesville: $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E}$.

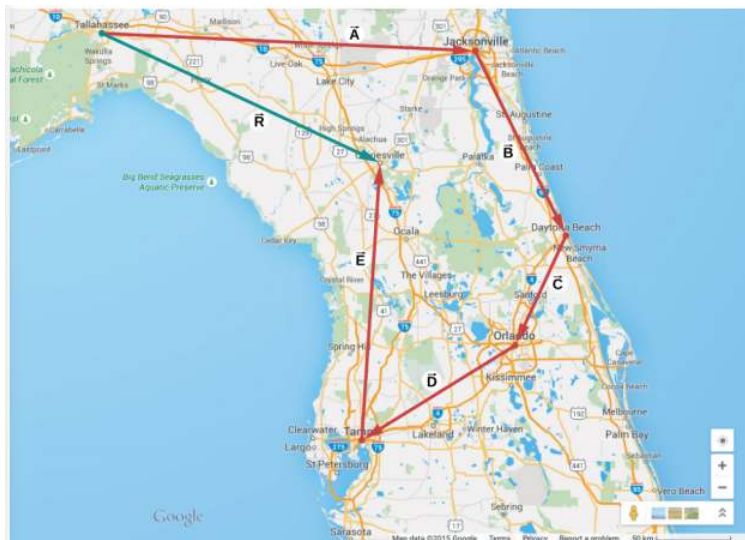


Figure 2.1.4: When we use the parallelogram rule four times, we obtain the resultant vector $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E}$, which is the green vector connecting Tallahassee with Gainesville.

Drawing the resultant vector of many vectors can be generalized by using the following tail-to-head geometric construction. Suppose we want to draw the resultant vector $\vec{R}$ of four vectors $\vec{A}$, $\vec{B}$, $\vec{C}$, and $\vec{D}$ (Figure 2.1.5a). We select any one of the vectors as the first vector and make a parallel translation of a second vector to a position where the origin (“tail”) of the second vector coincides with the end (“head”) of the first vector. Then, we select a third vector and make a parallel translation of the third vector to a position where the origin of the third vector coincides with the end of the second vector. We repeat this procedure until all the vectors are in a head-to-tail arrangement like the one shown in Figure 2.1.5. We draw the resultant vector $\vec{R}$ by connecting the origin (“tail”) of the first vector with the end (“head”) of the last vector. The end of the resultant vector is at the end of the last vector. Because the addition of vectors is associative and commutative, we obtain the same resultant vector regardless of which vector we choose to be first, second, third, or fourth in this construction.

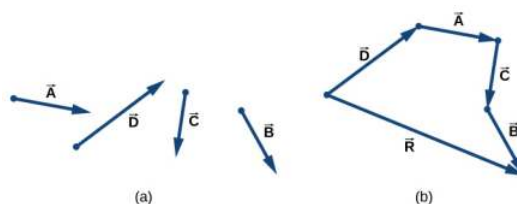


Figure 2.1.5: Tail-to-head method for drawing the resultant vector $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$. (a) Four vectors of different magnitudes and directions. (b) Vectors in (a) are translated to new positions where the origin (“tail”) of one vector is at the end (“head”) of another vector. The resultant vector is drawn from the origin (“tail”) of the first vector to the end (“head”) of the last vector in this arrangement.

✓ Example 2.1.2: Geometric Construction of the Resultant

The three displacement vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ in Figure 2.1.6 are specified by their magnitudes $A = 10.0$, $B = 7.0$, and $C = 8.0$, respectively, and by their respective direction angles with the horizontal direction $\alpha = 35^\circ$, $\beta = -110^\circ$, and $\gamma = 30^\circ$. The physical units of the magnitudes are centimeters. Choose a convenient scale and use a ruler and a protractor to find the following vector sums: (a) $\vec{R} = \vec{A} + \vec{B}$, (b) $\vec{D} = \vec{A} - \vec{B}$, and (c) $\vec{S} = \vec{A} - 3\vec{B} + \vec{C}$.

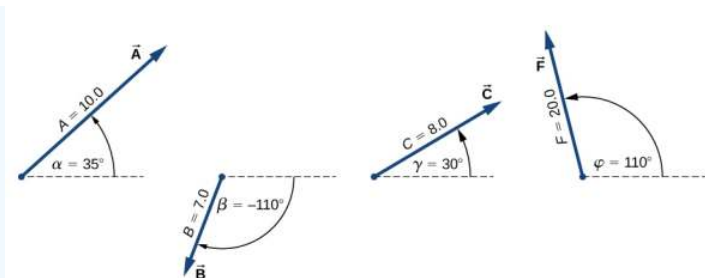


Figure 2.1.6: Vectors used in Example 2.1.2 and in the Exercise feature that follows.

Strategy

In geometric construction, to find a vector means to find its magnitude and its direction angle with the horizontal direction. The strategy is to draw to scale the vectors that appear on the right-hand side of the equation and construct the resultant vector. Then, use a ruler and a protractor to read the magnitude of the resultant and the direction angle. For parts (a) and (b) we use the parallelogram rule. For (c) we use the tail-to-head method.

Solution

For parts (a) and (b), we attach the origin of vector $\vec{B}$ to the origin of vector $\vec{A}$, as shown in Figure 2.1.7, and construct a parallelogram. The shorter diagonal of this parallelogram is the sum $\vec{A} + \vec{B}$. The longer of the diagonals is the difference $\vec{A} - \vec{B}$. We use a ruler to measure the lengths of the diagonals, and a protractor to measure the angles with the horizontal. For the resultant $\vec{R}$, we obtain $R = 5.8$ cm and $\theta_R \approx 0^\circ$. For the difference $\vec{D}$, we obtain $D = 16.2$ cm and $\theta_D = 49.3^\circ$, which are shown in Figure 2.1.7.

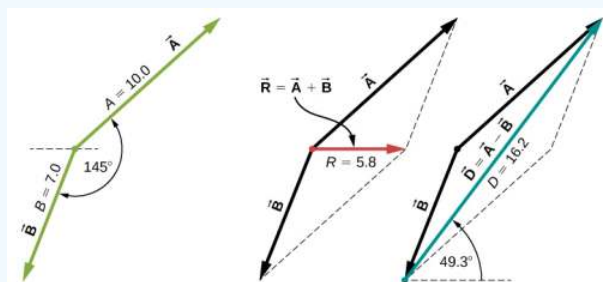


Figure 2.1.7: Using the parallelogram rule to solve (a) (finding the resultant, red) and (b) (finding the difference, blue).

For (c), we can start with vector $-3\vec{B}$ and draw the remaining vectors tail-to-head as shown in Figure 2.1.8. In vector addition, the order in which we draw the vectors is unimportant, but drawing the vectors to scale is very important. Next, we draw vector $\vec{S}$ from the origin of the first vector to the end of the last vector and place the arrowhead at the end of $\vec{S}$. We use a ruler to measure the length of $\vec{S}$, and find that its magnitude is $S = 36.9$ cm. We use a protractor and find that its direction angle is $\theta_S = 52.9^\circ$. This solution is shown in Figure 2.1.8.

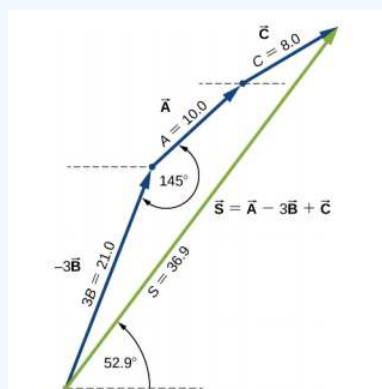


Figure 2.1.8: Using the tail-to-head method to solve (c) (finding vector $\vec{S}$, green).

? Exercise 2.3

Using the three displacement vectors $\vec{A}$, $\vec{B}$, and $\vec{F}$ in Figure 2.1.6, choose a convenient scale, and use a ruler and a protractor to find vector $\vec{G}$ given by the vector equation $\vec{G} = \vec{A} + 2\vec{B} - \vec{F}$.

📌 Simulation

Observe the addition of vectors in a plane by visiting this [vector calculator](#) and this [PhET simulation](#).

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2.2: Graphical Methods of Vector Addition

 This man shooting a basketball requires a subconscious understanding of vectors

Figure 2.2.1: Copy and Paste Caption here. (Copyright; author via source)

Successfully shooting a basketball requires a subconscious understanding of the vectors involved in how the basketball moves through the air. The vertical and horizontal vectors must be perfectly organized if the ball is to pass through the basket.

Graphical Methods Vector Addition

In physics, a quantity, such as mass, length, or speed that is completely specified by its magnitude and has no direction is called a **scalar**. A **vector**, on the other hand, is a quantity possessing both magnitude and direction. A vector quantity is typically represented by an arrow-tipped line segment. The length of the line, drawn to scale, represents the magnitude of the quantity. The direction of the arrow indicates the direction of the vector. Not only can vectors be represented graphically, but they can also be added graphically.

For one dimensional **vector addition**, the first vector is placed on a number line with the tail of the vector on the origin. The second vector is placed with its tail exactly on the arrow head of the first vector. The sum of the two vectors is the vector that begins at the origin and ends at the arrow head of the final added vector.

Consider the following two vectors.

 The top number line shows a vector with magnitude 11 in the positive direction and the bottom number line shows a vector with magnitude -3

Figure 2.2.2

The red vector has a magnitude of 11 in the positive direction on the number line. The blue vector has a magnitude of -3, indicating 3 units in the negative direction on the number line. In order to add these two vectors, we place one of the vectors on a number line and then the second vector is placed on the same number line such that its origin is on the arrow head of the first vector.

 One number line with one vector with magnitude 11 in the positive direction and another vector overlapping with magnitude -3

Figure 2.2.3

The sum of these two vectors is the vector that begins at the origin of the first vector (the red one) and ends at the arrow head of the second (blue) vector. So the sum of these two vectors is the purple vector, as shown below.

 One number line with one vector with magnitude 11 in the positive direction and another vector overlapping with magnitude -3, showing the sum of the two vectors with a third with magnitude 8

Figure 2.2.4

The vector sum of the first two vectors is a vector that begins at the origin and has a magnitude of 8 units in the positive direction. If we were adding three or four vectors all in one dimension, we would continue to place them head to toe in sequence on the number line. The sum would be the vector that begins at the beginning of the first vector and goes to the ending of the final vector.

Adding Vectors in Two Dimensions

In the following image, vectors A and B represent the two displacements of a person who walked 90. m east and then 50. m north. We want to add these two vectors to get the vector sum of the two movements.

 Graph of two vectors, representing two displacements of a person who walked 90 mi east and then 50 mi north

Figure 2.2.5: Copy and Paste Caption here. (Copyright; author via source)

The graphical process for adding vectors in two dimensions is to place the tail of the second vector on the arrow head of the first vector as shown above.

The sum of the two vectors is the vector that begins at the origin of the first vector and goes to the ending of the second vector, as shown below.

 Two vectors representing two displacements of a person who walked 90 mi east and then 50 mi north, with the third vector connecting the two, creating a triangle indicating the sum

Figure 2.2.6: Copy and Paste Caption here. (Copyright; author via source)

If we are using totally graphic means of adding these vectors, the magnitude of the sum would be determined by measuring the length of the sum vector and comparing it to the original standard. We would also use a compass to measure the angle of the summation vector.

If we are using calculation means, we can divide 50. m by 90. m and determine inverse tangent of the dividend. The result of 29.05 indicates the angle of 29° north of east. The length of the sum vector can also be determined mathematically by the Pythagorean theorem, $a^2 + b^2 = c^2$. In this case, the length of the hypotenuse would be the square root of (8100 + 2500) or 103 m.

If three or four vectors are to be added by graphical means, we would continue to place each new vector head to toe with the vectors to be added until all the vectors were in the coordinate system. The sum vector is the vector from the origin of the first vector to the arrowhead of the last vector. The magnitude and direction of the sum vector can be measured.



Figure 2.2.7: Copy and Paste Caption here. (Copyright; author via source)

Have you ever used a phone app that provides directions or a navigation system in your car? These programs help you get from Point A to Point B by breaking it down into a series of left and right turns that exemplify many of the graphical methods of vector addition described above. The navigation systems in self-driving cars are even more advanced. Continue to practice vector addition by helping a driverless car get to its destination in the following simulation:

Summary

- Scalars are quantities, such as mass, length, or speed, that are completely specified by magnitude and have no direction.
- Vectors are quantities possessing both magnitude and direction and can be represented by an arrow; the direction of the arrow indicates the direction of the quantity and the length of the arrow is proportional to the magnitude.
- Vectors that are in one dimension can be added arithmetically.
- Vectors that are in two dimensions are added geometrically.
- When vectors are added graphically, graphs must be done to scale and answers are only as accurate as the graphing.

Review

1. On the following number line, add the vector 7.5 m/s and the vector -2.0 m/s.

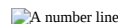
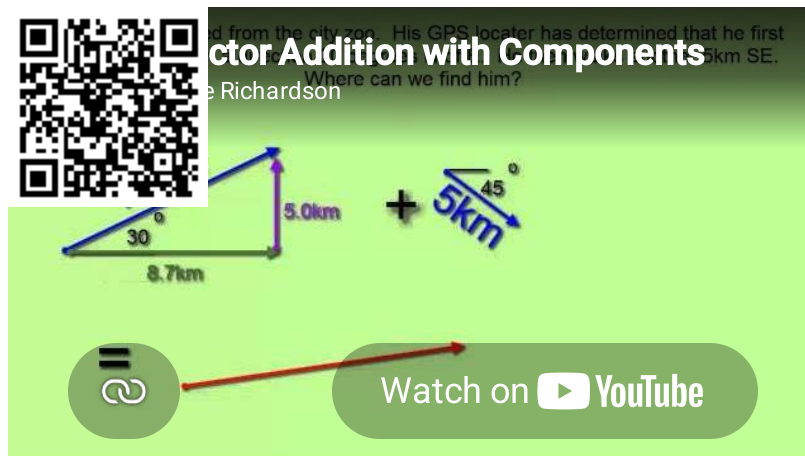


Figure 2.2.8

2. On a sheet of graph paper, add a vector that is 4.0 km due east and a vector that is 3.0 km due north.

Explore More

Use this resource to answer the questions that follow.



The image is a video thumbnail for a resource titled "Vector Addition with Components". It features a QR code in the top left corner. The main content shows a vector diagram with a resultant vector of 8.7 km at a 30° angle, and two component vectors: one of 5.0 km pointing vertically upwards and another of 5 km pointing at a 45° angle. Text overlays include "d from the city zone. His GPS locator has determined that he first 5km SE. Where can we find him?" and "e Richardson". At the bottom, there is a "Watch on YouTube" button with a play icon.

1. What is a resultant?
2. What are the steps necessary to add vectors in two dimensions?

Additional Resources

[Real World Application: Drift](#)

[Real World Application: Threading the Needle](#)

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